



Fig. 4: Applications of an external OFS



Fig. 5: An Interior OFS

factor that has a high impact on road safety, which is visibility. This vital information is needed for optimising road maintenance and for making decisions regarding speed limits using variable signs.

OFSes, measuring atmospheric visibility (meteorological optical range) by determining the amount of light scattered by particles in the air, are widely used in ports and harbours.

Interior role. How would you know if you are maintaining a safe distance from vehicles around you when headlights cannot penetrate fog? The new class of compact, sensitive and reliable interior OFSes could come in handy.

The windshield-mountable interior OFS provides an output signal that corresponds to the level of moisture on the interior glass surface of the vehicle.

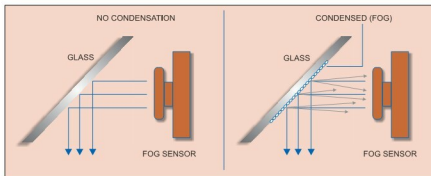


Fig. 6: Working principle of an interior OFS

Note that, unlike the common (exterior) OFS, the interior fog sensor works by transmitting a beam of infrared light onto the interior glass. When fog is present, the beam is scattered back to the sensor. This information is then used by the vehicle's climate control system to determine if the defogger or glass-heating elements should be activated.

The fog sensor's high sensitivity allows fog to be detected by the sensor before it is detected by the human eye. This not only improves driving safety but also the occupants' comfort and convenience. For instance, if a vehicle can decipher that it is foggy outside, it can automatically switch on fog lamps or emergency flashers, helping the driver see, along with helping other motorists notice the occupied vehicle.

In short, OFSes are suitable for a vast range of applications including, but not limited to:

1. Providing automatic control of fog sirens, navigation and warning lights
2. Weather forecasting. These could be installed on ocean-going data buoys, providing real-time fog and visibility data
3. Integrating into existing systems for synoptic weather observations

Hitting the mark

In order to get the best from an OFS (exterior or interior), you must do your home work well. The exterior sensor should be mounted such that it looks horizontal and there must not be anything in the sightline

closer than 5m to 10m. A simple shield protecting the sensor from direct sunlight and lenses from rain, reaching not more than 100mm in the laser beam direction, helps keep away precipitation from the optics. But in most cases, the existing small shield on the sensor front is sufficient.

When mounting an interior sensor (on the windshield), try and limit the overall distance between the glass and the sensor in the range of 4cm to 10cm. The best mounting position for this sensor is between the windshield and the rear-view mirror.

For DIYs

With the help of a ready-made exterior or interior OFS, you can build your own fog-related projects for your home, office and vehicle. Working voltage of an optical fog detector is between 6V DC and 16V DC, with a current requirement in the range of 60mA to 600 mA (refer official datasheet of your OFS).

If you want to study the atmospheric visibility of fog and its constituents, or if you want to fabricate a smart fog-triggered warning system for automobiles, an external MCU circuitry and a few lines of software are required to process output signals (serial data or analogue voltage) available from the OFS.

For basic projects, string output from the OFS can be routed to a logger with RS232 input or to a PC via the serial port and with a terminal program like Hyper Terminal. ●